31. Which of the following statement is not true of two genes that show 50% recombination frequency? (NEET 2013)
a) The genes may be on different chromosomes
b) **The genes are tightly linked**
c) The genes show independent assortment
d) If the genes are present on the same chromosome, they undergo more than one crossover in every meiosis.

32. Haploids are more suitable for mutation studies than the diploids. This is because (AIPMT 2008)
a) **All mutations, whether dominant or recessive are expressed in haploids**
b) Haploids are reproductively more stable than diploids
c) Mutagens penetrate in haploids more effectively than diploids
d) Haploids are more abundant in nature than diploids

33. Crossing over that results in genetic recombination in higher organisms occurs between (AIPMT 2004)
a) **Non-sister chromatids of a bivalent**
b) Two daughter nuclei
c) Two different bivalents
d) Sister chromatids of bivalents

34. Removal of introns and joining the exons in a defined order in a transcription unit is called (AIPMT 2009, AIPMT Pre 2012)
a) Tailing b) Transformation
c) Capping d) **Splicing**

35. Selection the correct option (AIPMT 2014)\

	Direction of RNA synthesis	**Direction of reading of the template DNA strand**
a)	**5' – 3'**	**3' – 5'**
b)	3' – 5'	5' – 3'
c)	5' – 3'	5' – 3'
d)	3' – 5'	3 ' – 5'

36. Peptide synthesis inside a cell takes place in (AIPMT 2011)
a) **Ribosomes** b) Chloroplast
c) Mitrochondria d) Chloroplast

37. During protein synthesis in a organism at one point the process comes to a halt. Select the group of the three codons from the following from which any one of the three could bring about this halt. (AIIMS 2006)
a) UUU, UCC, UAU b) UUUC, UUA, UAC
c) **UAG, UGA, UAA** d) UUG, UCA, UCG

38. The binding site of tRNA with mRNA and amino acids respectively are (AIIMS 2009)
a) mRNA with DHU loop and amino acid with CCA end
b) mRNA with CCA end and amino acid with anticodon loop
c) mRNA with anticodon loop and amino acid with DHU loop
d) **mRNA with anticodon loop and amino acid with CCA end**

39. Which of the following is correct regarding genetic code? (AIIMS 2010)
a) UUU is the initiation codon which also codes for phenylalanine
b) **There are 64 triplet codons and only 20 amino acids**
c) Three random nitrogen bases specify the placement of one amino acid
d) UAA is the nonsense codon which also codes for methionine

40. Which of the following set of options is used in translation? (AIIMS 2015)
a) hnRNA, tRNA, rRNA
b) **mRNA, tRNA, rRNA**
c) mRNA, tRNA, hnRNA
d) hnRNA, rRNA, lRNA

41. Sequence of DNA (non-coding) is known as (JIPMER 2006)
a) exon b) **intron**
b) cistron d) none of these

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